Project planning report

Saved project inputs and selected calculation results

2026-10-27 22:30:25.626741+00:00

Project

This report presents saved project inputs and the original results of the selected calculations. Country-specific requirements, electrical design and installation checks remain open. It does not approve the complete installation.

Project	Linker production acceptance · owned synthetic project
Project reference	1812286480344d6d8cec94e2f5f2c00e
Input revision	12

Site and project inputs

Location	Paris · public city-centre reference · synthetic acceptance
Country	FR
Time zone	Europe/Paris
Project note	Owned synthetic production acceptance. No customer data, no installation claim, no inquiry.

Provisional bill of materials

This list combines the selected equipment, saved roof modules and documented included components. Quantities are shown where established. Mounting, cabling, protection and other installation materials still require project-specific design.

Quantity	Product	Reference	Supply scope
20 pc	Trina Solar TSM-460NEG9R.28		PV reference module · project quotation

PV modules used in the calculation

Module counts come from the saved layout of the selected generation result. The listed modules are design references; electrical and mounting suitability require project-specific assessment.

Input revision	11
PV nameplate power	9.2 kWp
Source document · Page 2	Trina Solar TSM-460NEG9R.28 https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf

Quantity	Model	Module power (STC)
20	Trina Solar TSM-460NEG9R.28	460 W

PV generation scenario

This section presents the selected saved PV profile and its original assumptions. Detailed shading and electrical design require separate assessment.

PV generation	9,315.201 kWh
Calculation period (UTC)	2023-01-01T00:00:00+00:00 – 2024-01-01T00:00:00+00:00
Calculated intervals	8,760

Weather-based PV reference: sources and assumptions

This unshaded reference uses typical weather assembled from the selected source months. Its synthetic calculation year defines only the output time axis. The calculated generation is neither measured generation nor a forecast for that year. Nearby and inter-row shading, real MPPT limits, device-specific inverter behavior and equipment or country approval are not assessed. The generic conversion model uses only the stated parameters; no additional PVGIS system-loss percentage is applied. The AC total shown covers the complete retained PV reference period. The separate energy calculation uses its own period and AC or DC input selection. Array azimuth is measured clockwise from north.

The stated SAPM thermal coefficients are reference assumptions. Their applicability to the saved module clearance and actual ventilation has not been established. Source notes retain the original declared wording. Each source below belongs to its named calculation; different scenarios are not added together.

PV model / library version	uniform_unshaded_pvlib_reference_v1 / pvlib 0.16.1
Typical-weather source selection period	2014–2023
Radiation database	PVGIS-SARAH3
Meteorological database	ERA5
Terrain horizon	Omitted
Synthetic energy-axis year	2023
Calculation period (UTC)	2023-01-01T00:00:00+00:00 – 2024-01-01T00:00:00+00:00
Calculated intervals	8760
Modelled AC generation	9315.200808684325 kWh
Original irradiance time offset (hours / microseconds)	0.1805 h / 649800000 µs
Selected source month: year (January to December)	1: 2014; 2: 2015; 3: 2023; 4: 2019; 5: 2016; 6: 2023; 7: 2016; 8: 2023; 9: 2016; 10: 2016; 11: 2018; 12: 2018
PV interval reconstruction policy	representative_pv_output_over_declared_intervals_v1
Solar-position date basis	original_selected_year_plus_reported_irradiance_offset
Native radiation time semantics	sarah_instantaneous
Original official JRC weather request	https://re.jrc.ec.europa.eu/api/v5_3/tmy?lat=48.8566&lon=2.3522&startyear=2014&endyear=2023&usehorizon=0&outputformat=json https://re.jrc.ec.europa.eu/api/v5_3/tmy?lat=48.8566&lon=2.3522&startyear=2014&endyear=2023&usehorizon=0&outputformat=json
Original response received (UTC)	2026-10-27T19:31:38.475826+00:00
PV nameplate power	9.2 kWp
Original reference assumption or applicability note	Explicit unshaded reference using the selected native Paris TMY (2014-2023 selection, synthetic 2023 UTC axis), current Woo-maintained Trina module evidence and the saved synthetic 20-module roof. The generic conversion and thermal parameters do not establish real equipment performance, mounting fit, electrical/installation qualification or country/grid approval. Existing 2024 load profiles, imported PV profiles and original results are preserved and are not remapped or adopted.
Source calculation / input revision	f32e226aa7d74ae9a65e655dffb4cc22 / 12 / pv_available_ac
Source calculation / input revision	e9875888ea9d4273ae162448554de1ab / 11 / pv_available_ac
Included modules: quantity × model / module STC; document SHA-256 / page	20 × Trina Solar TSM-460NEG9R.28 / 460.0 W; f4d280f9a0d8947 5cf575fbb99906e2b091da27f08b848fc51877cc1d24e8e22 / p. 2
Array identity	array_b4e6aac277b644bbad5eaece725b993f
Array: module count / STC / tilt / azimuth clockwise from north / ENU normal	20 / 9200.0 W / 10.0° / 0.0° / [0.0, 0.17364817766693033, 0.984807753012208]
Saved module lower-edge clearance	0.15 m

Module power temperature coefficient	-0.0029 / °C
Transposition model / albedo	perez / 0.2
Incidence modifier: model / beam b0 / sky diffuse / ground diffuse	ashrae / 0.0 / 1.0 / 1.0
Retained fractions: spectral / soiling / DC wiring / DC mismatch	1.0 / 1.0 / 1.0 / 1.0
Thermal model: a / b (s/m) / delta T (°C)	sapm_cell; a=-3.47; b=-0.0594 s/m; delta T=3.0 °C
Original reference assumption or applicability note	Declared reference: Perez, albedo 0.2, ASHRAE b=0, and all diffuse/spectral/soiling/wiring/mismatch retention factors equal to 1. Near-field and inter-row shading are not modelled; the selected weather also omits terrain horizon. These are scenario assumptions, not measured losses or shading approval.
Original reference assumption or applicability note	pvlib 0.16.1 SAPM open_rack_glass_glass: a=-3.47, b=-0.0594, deltaT=3. Declared reference mounting, no site calibration. Suitability of this open-rack preset for the saved 0.15 m lower-edge clearance and actual ventilation is not established.
Original reference assumption or applicability note	Synthetic mounting assumptions for acceptance only; no structural, shading or site approval.
Generic converter identity / assigned arrays	converter_75a7239e5e9644859dc8f10e37a66cf8 / ['array_b4e6aac277b644bbad5eaece725b993f']
Conversion model / DC input assumption	pvwatts / independent_mppt_array_mpp_sum
Converter reference DC power / modelled AC cap	10416.666666666666 W / 9999.999999999998 W
Conversion efficiency: nominal / reference	0.96 / 0.9637
Original reference assumption or applicability note	NREL PVWatts reference conversion: eta_nom=0.96, eta_ref=0.9637, DC reference=10416.666666666666 W and AC reference=10 kW. No selected inverter SKU or manufacturer curve. No JRC 14% system loss is added to this separate weather-based model.
Frozen source artifact / SHA-256	geometry: 7fb6636f01ca3e47352f5528f9851125b90d528a76a425cb0cc0f176ce4bf807
Frozen source artifact / SHA-256	pv_model_inputs: 2dcfd9ee1501d537a173efe7f3dff143a4a39e9bcf108fc0a9383b314d10047b
Frozen source artifact / SHA-256	pv_model_witness: 12ac831be719340e4d8a843587a5b8fa2e09b1cea8cc6eb1cdd3f8e92325a1bc
Frozen source artifact / SHA-256	weather_capture: e1182b9d7b6cee37ea8946301a6a656517f6bc9328e3e19aedb1843ba159ae30
Frozen source artifact / SHA-256	weather_original: c0ec4d04ad4293a1e008e7b09919dac75a5dbd39a4f81b4a3c77d7a5245790e5

Energy scenario

These values come from the selected saved energy result and reflect its original scenario parameters. The general energy-flow model does not establish the performance of the selected equipment.

Calculation period (UTC)	2023-06-16T09:00:00+00:00 – 2023-06-16T11:00:00+00:00
Simulated load demand	1.103 kWh
PV generation	12.341 kWh
Grid import	0 kWh
PV export	11.238 kWh
PV curtailment	0 kWh
Unserved simulated load	0 kWh
Solar share of gross demand	100 %

PV used or stored on site	8.934 %
Calculated intervals	2

Saved calculation references

Data source	Synthetic 2023 zero external AC auxiliary load · Paris reference scenario
Data source	Owned synthetic PV/load-only baseline: explicitly zero storage capacity and power, unit efficiencies, zero loss. No SKU performance assertion.
Data source	Synthetic 2023 gross load · Paris reference scenario
Data source	JRC PVGIS typical meteorological year
Calculation reference	f32e226aa7d74ae9a65e655dffb4cc22 · Input revision 12
Calculation reference	e9875888ea9d4273ae162448554de1ab · Input revision 11